

Structural interpretation of emergency unemployment assistance

Model 15: mechanisms, calibration, weaknesses and a submission-ready research agenda

Labour economics teaching deck | Based on manuscript v10 and structural model 15 | August 2026



What this deck is designed to teach

Read the mechanism

Translate benefits, work wedges and matching conditions into search, acceptance and employment exits.

Interrogate identification

See which moments discipline which parameters—and why near-equivalent fits do not identify a unique mechanism.

Use the model honestly

Separate a useful quantitative interpretation exercise from welfare analysis or a fully structural equilibrium model.

The goal is not to defend every current modelling choice; it is to show what can be learned now and what must change before journal submission.

Bottom line on model 15

Real advance

Model 15 gives bounded, surplus-sensitive exit probabilities; shares a common surplus index across finding and exits; and genuinely holds out Australian post JFR.

Main empirical message

In the benchmark, the benefit-only job-finding DiD is -1.50 pp versus -1.31 pp in the full environment. For exits, nonlinear interaction is central.

Binding limitation

The selected benchmark is weakly identified. Near-fitting parameterisations imply held-out errors from roughly -2.84 to $+0.48$ pp and work wedges from 5% to 41% of wages.

Treat the current model as a partially identified interpretation model. Do not present its point decomposition as uniquely estimated.

Roadmap

Part I — Learn the current model

1. Empirical moments
2. Worker problem and search
3. Acceptance and exit hazards
4. Firms, matching and dynamics
5. Calibration mapping

Part II — Decide what to keep

6. Benchmark results
7. Identification audit
8. Internal-consistency audit
9. Minimal submission-ready model
10. Ambitious extensions and paper write-up

Slides deliberately alternate equations, intuition, diagnostics and recommended decisions.

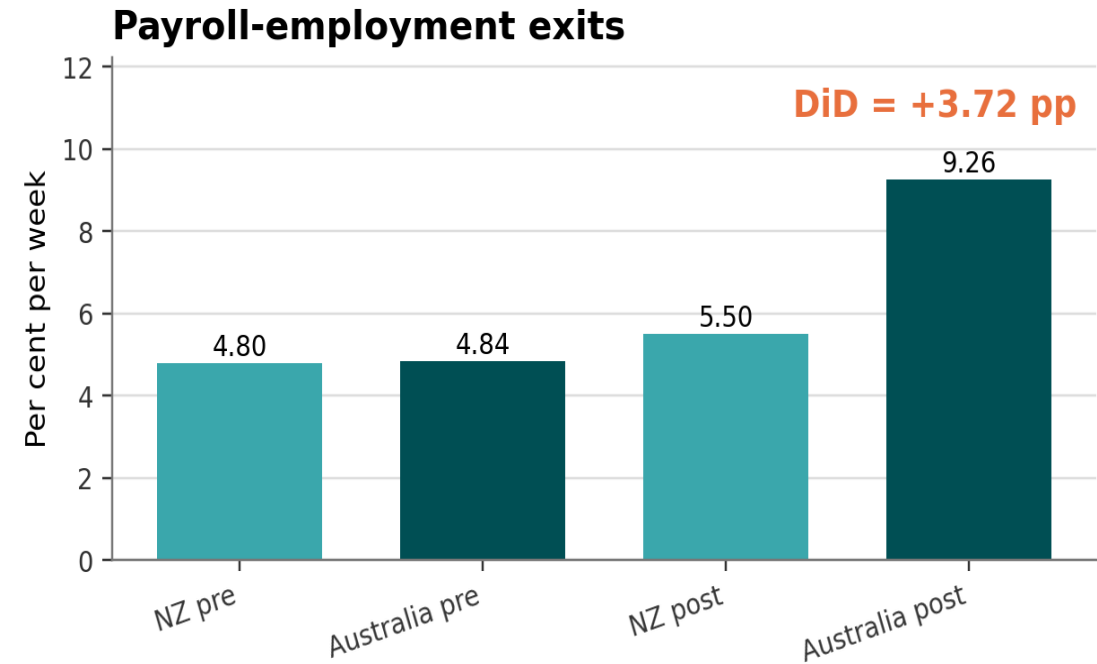
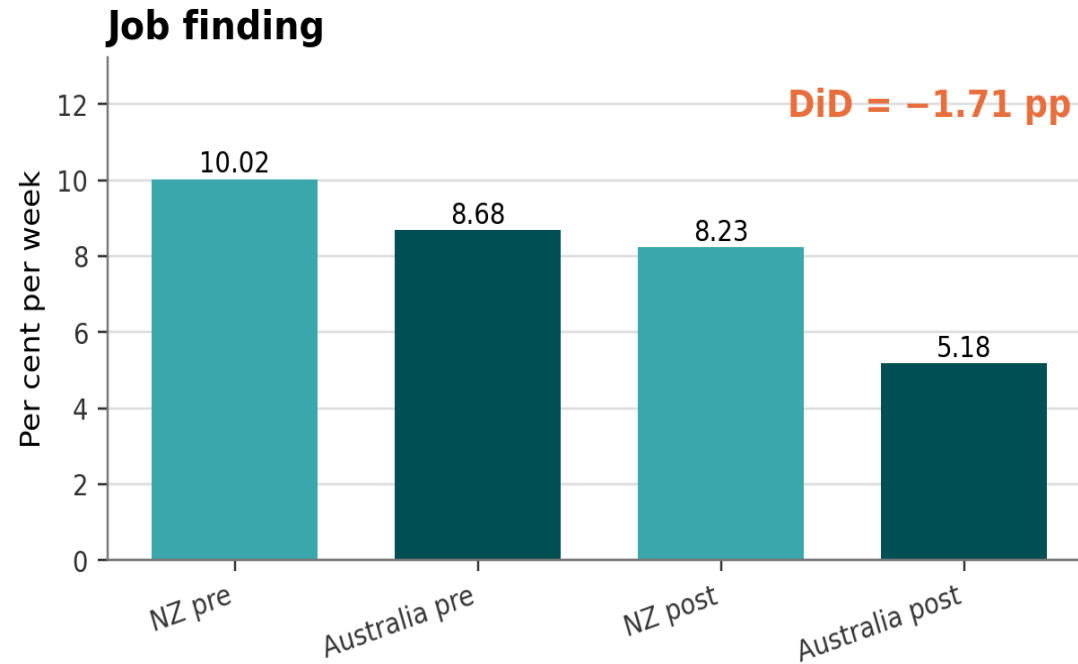
01

What must the model explain?

Start from transition moments; ask which mechanisms can rationalise them.

The draft supplies four transition cells—and two DiD contrasts

The draft's four empirical cells discipline model 15



Source: manuscript Tables 3–4 / model-15 target vector. Percentage-point effects use Australian minus New Zealand changes.

These numbers are provisional until the empirical risk sets and inference are rebuilt.

Why add structure to a credible reduced-form design?

Reduced form can estimate

- The change in Australian-minus-NZ job finding
- The change in Australian-minus-NZ payroll exits
- Heterogeneity by predicted policy exposure
- Dynamic responses around announcements

Structure can organise

- Benefits versus common job-opportunity shocks
- Search versus acceptance margins
- Complementarity in match continuation
- Counterfactual changes outside the observed episode

Structure does not repair a selected empirical sample. Revised target moments must come first.

What changed from the manuscript model to version 15?

Dimension	Manuscript / earlier model	Model 15	Implication
Employment exits	Largely exogenous or additively shifted	Smooth, bounded hazard of net flow surplus	Benefits and work wedges interact nonlinearly
Post calibration	Fits both groups' post JFRs	NZ post JFR fitted; AU post JFR held out	Creates a genuine validation target
Search costs	May vary pre/post	Restricted model holds κ fixed	Common post shock is channelled through μ
Decomposition	Narrative «health versus benefits»	Four scenario environments	Scenarios are not additive contributions
Exit language	Match-quality interpretation	Static group-level logistic hazard	Must be described as reduced-form unless quality state is implemented

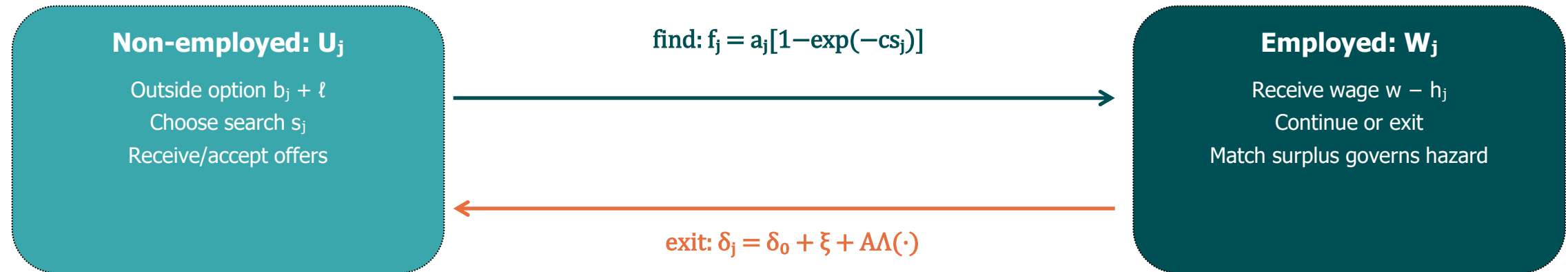
The paper's current structural section and abstract must be rewritten from model 15 outputs—not patched line by line.

02

Model anatomy

Two types, two states, a common market, and a surplus-sensitive exit hazard.

The model has two worker types and two labour-market states



$j \in \{R, N\}$: R is benefit-eligible / Australian; N is ineligible / New Zealand comparison.

This is a partial-equilibrium interpretation model. The two types share job opportunities, but their outside options and residual work wedges differ.

Flow payoffs create the net work-surplus index

Eligible / Australian type R

Outside: $b^R + \ell$

Work: $w - h^R$

Flow surplus: $z^R = w - h^R - b^R - \ell$

The supplement raises b^R and lowers z^R .

Ineligible / NZ type N

Outside: $b^N + \ell$, with $b^N = 0$ in code

Work: $w - h^N$

Flow surplus: $z^N = w - h^N - b^N - \ell$

Normalisation sets $h^N = 0$ post-shock.

The code labels h as «health»; empirically it is a residual differential net work-surplus wedge that may absorb composition, policy interactions and measurement.

A job is found only if an offer arrives and is accepted

Offer arrival

$$o(s,c) = 1 - \exp(-cs)$$

c is the contact probability from the matching block; s is chosen search effort. The exponential form keeps offers in $[0,1]$.

Acceptance

$$a(z) = \Lambda[(z - \bar{z})/\sigma_a]$$

Job-finding probability:

$$f = a(z) \times o(s,c).$$

Threshold $\bar{z}$ and dispersion σ_a are fixed, not estimated.

Observed job finding is a product of search, contacts and acceptance. Aggregate transition rates cannot separately identify all three.

Search effort solves a dynamic marginal-cost condition

First-order condition

$$\kappa s^\eta = \beta \cdot a \cdot c \cdot \exp(-cs) \cdot [W' - U']$$

Marginal cost rises with curvature η ; the gain depends on acceptance, contact and next-period value surplus.

Interpretation

Higher benefits reduce $W-U$ and acceptance; both lower the return to search. Poor matching lowers c . Anticipated policy expiry changes $W'-U'$ before the cash payment ends.

In the selected calibration $\eta = 100$. That pins s close to one, so the model's «labour-supply» response is mostly an acceptance response.

Employment exits are a smooth hazard of static flow surplus

Implemented equation

$$\delta_j = \text{clamp}[\delta_0 + \xi + A \wedge((\bar{z}_s - z_j)/\sigma_s)]$$

δ_0 : baseline

ξ : common post shock

A : endogenous component amplitude

Economic meaning

A lower z_j moves a group into the steep region of the hazard. Benefits and h^R therefore complement each other: the effect of one depends on the level of the other.

There is no explicit match-quality state or joint continuation rule. Near term: call this a smooth reduced-form employment-exit hazard.

The matching block turns μ into contact opportunities

Current firm side

Firm surplus = $y - w$
 $J = (y - w) / [1 - \beta(1 - \delta_{\text{firm}})]$
 $\theta = [\beta \mu J / \text{vacancy cost}]^{(1/\alpha)}$
 $c = \mu \theta^{(1-\alpha)}$

Internal inconsistency

Firm continuation uses $\delta_{\text{firm}} = 1\%$ weekly, while worker-side exits are about 4.8%–9.3%. Policy-induced exits do not reduce J or feed back into vacancy creation.

Choose: integrate the same exit hazard into firm value, or treat c/μ as an exogenous opportunity process and drop free-entry language.

Bellman equations make the shock dynamic

Employed value

$$W_{j,t} = (w - h_{j,t}) + \beta[(1 - \delta_{j,t})W_{j,t+1} + \delta_{j,t} U_{j,t+1}]$$

Non-employed value

$$U_{j,t} = (b_{j,t} + \ell) - C(s_{j,t}) + \beta[f_{j,t} W_{j,t+1} + (1 - f_{j,t})U_{j,t+1}]$$

Backward induction solves values and search over the announced shock horizon; forward simulation averages weeks 12–36.

Timing assumptions are economically important

- Weekly model: $T = 60$; shock weeks 12–36 inclusive (25 weeks).
- Shock is unanticipated at onset, then its end date is known with certainty.
- The manuscript's empirical window, announcement date and supplement payment window do not currently line up with the model averaging window.
- $\beta = 0.99$ weekly implies an annual discount factor near 0.59—far below conventional calibrations.
- The real policy was announced, paid, extended and tapered in stages; expectations should follow that history.

Create a date-to-model-week table and set $\beta_{\text{week}} = \beta_{\text{annual}}^{(1/52)}$. Re-run every quantitative result.

03

Calibration and what the model says

A close fit can coexist with severe mechanism uncertainty.

Calibration is sequential: each empirical cell is assigned a job



Teaching point: «held out» is useful only if specification choices were not tuned after seeing the held-out result.

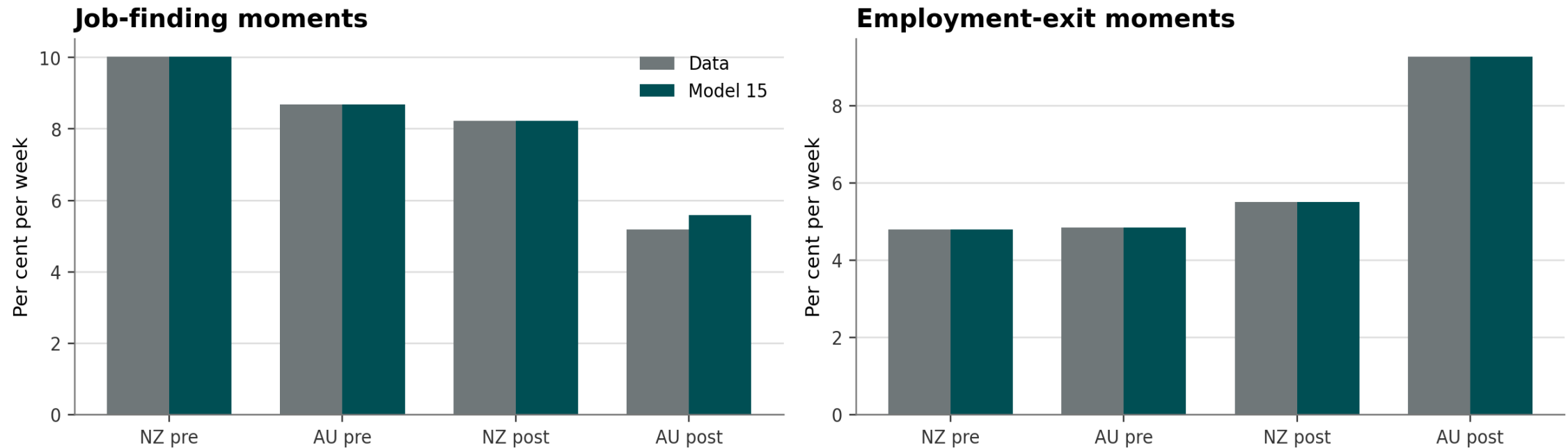
Core fixed and calibrated quantities in the benchmark

Object	Benchmark	How set	Interpretation / concern
Weekly β	0.99	Fixed	Implies implausibly low annual discounting
Wage / output	0.55 / 1.00	Fixed normalisation	No wage bargaining or wage distribution
Benefit RR	0.28 $\rightarrow$ 0.56	Stylised doubling	Actual policy is a flat \$550/fortnight plus means tests
Search curvature η	100	Selected from restricted grid	Pins search close to one
Exit threshold / dispersion	0.10 / 0.04	Selected from grid	Many near-ties have radically different predictions
μ pre / post	0.0739 / 0.0677	Fit JFR moments	Post μ fitted to NZ JFR
Differential wedge h^R	0.0368 = 6.69% wage	Fit AU post exits	Residual wedge, not identified health disutility

A manuscript calibration table should label every item fixed, externally calibrated, fitted or held out.

Benchmark fit: the held-out error is visible but modest

Fit is close by construction—except the held-out Australian post JFR

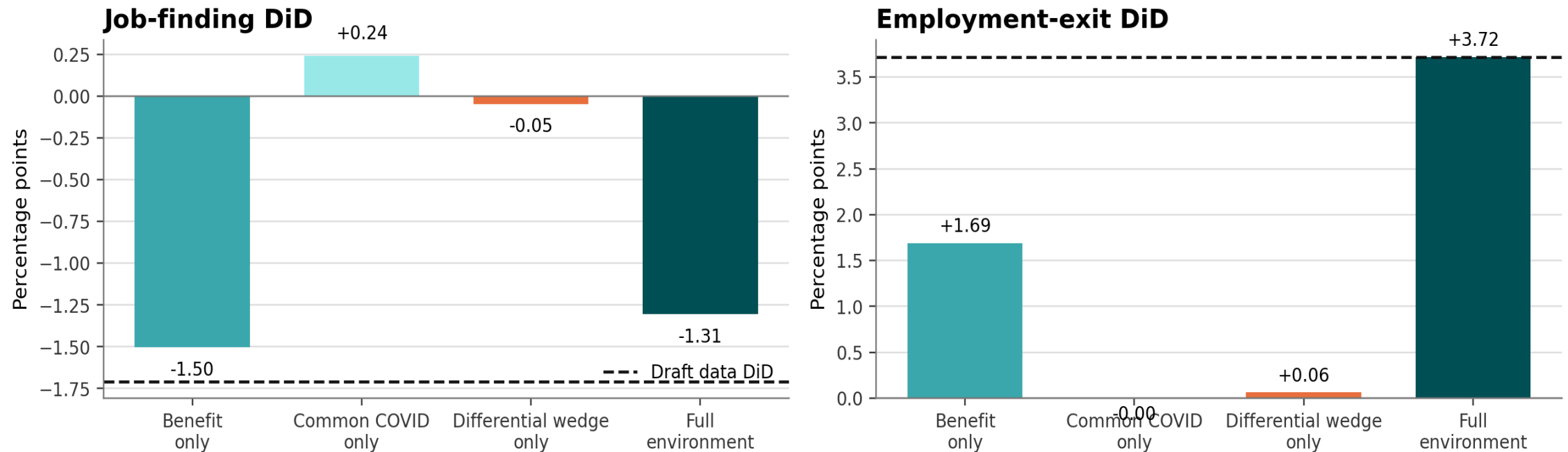


Benchmark predicts Australian post JFR 5.58% versus 5.18% in the draft (+0.40 pp). All four exit cells are calibration targets.

The fit bars are not standard errors; targets are treated as exact in the current code.

Scenario results: benefit effects and pandemic wedges interact

These are counterfactual environments—not additive shares



Nonlinear interactions matter. For exits, full DiD 3.72 pp exceeds the sum of standalone effects by about 1.97 pp.

Do not divide standalone bars by the full bar and call the result a «share».

The model reverses the manuscript's current job-finding headline

What the manuscript says

Most of the labour response came from pandemic work disutility rather than benefit generosity; benefit effects were smaller in normal conditions.

What model 15 says

Benefit-only job-finding DiD: -1.50 pp.
Full model DiD: -1.31 pp.
Common COVID only: $+0.24$ pp.
Differential wedge only: -0.05 pp.

**For job finding, the benefit channel is at least as large in the no-COVID counterfactual as in the full benchmark.
Rewrite abstract, introduction and conclusion.**

For exits, complementarity—not «health versus benefits»—is the result

Standalone DiD effects

Benefit only: +1.688 pp

Common COVID only: ≈ 0.000 pp

Differential wedge only: +0.061 pp

Sum: +1.749 pp

Full environment

Full exit DiD: +3.720 pp

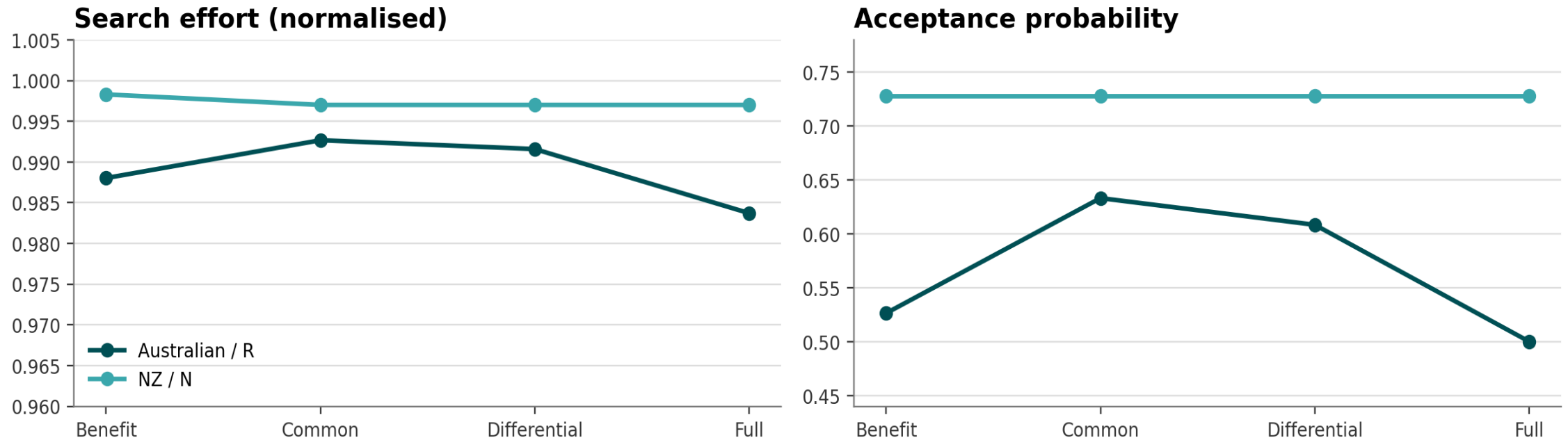
Residual interaction: $3.720 - 1.749 = +1.971$ pp

The interaction exceeds every standalone wedge other than benefits.

Compute all 2^3 counterfactual combinations and report a Shapley or explicit interaction decomposition.

The benchmark's job-finding mechanism is mainly acceptance

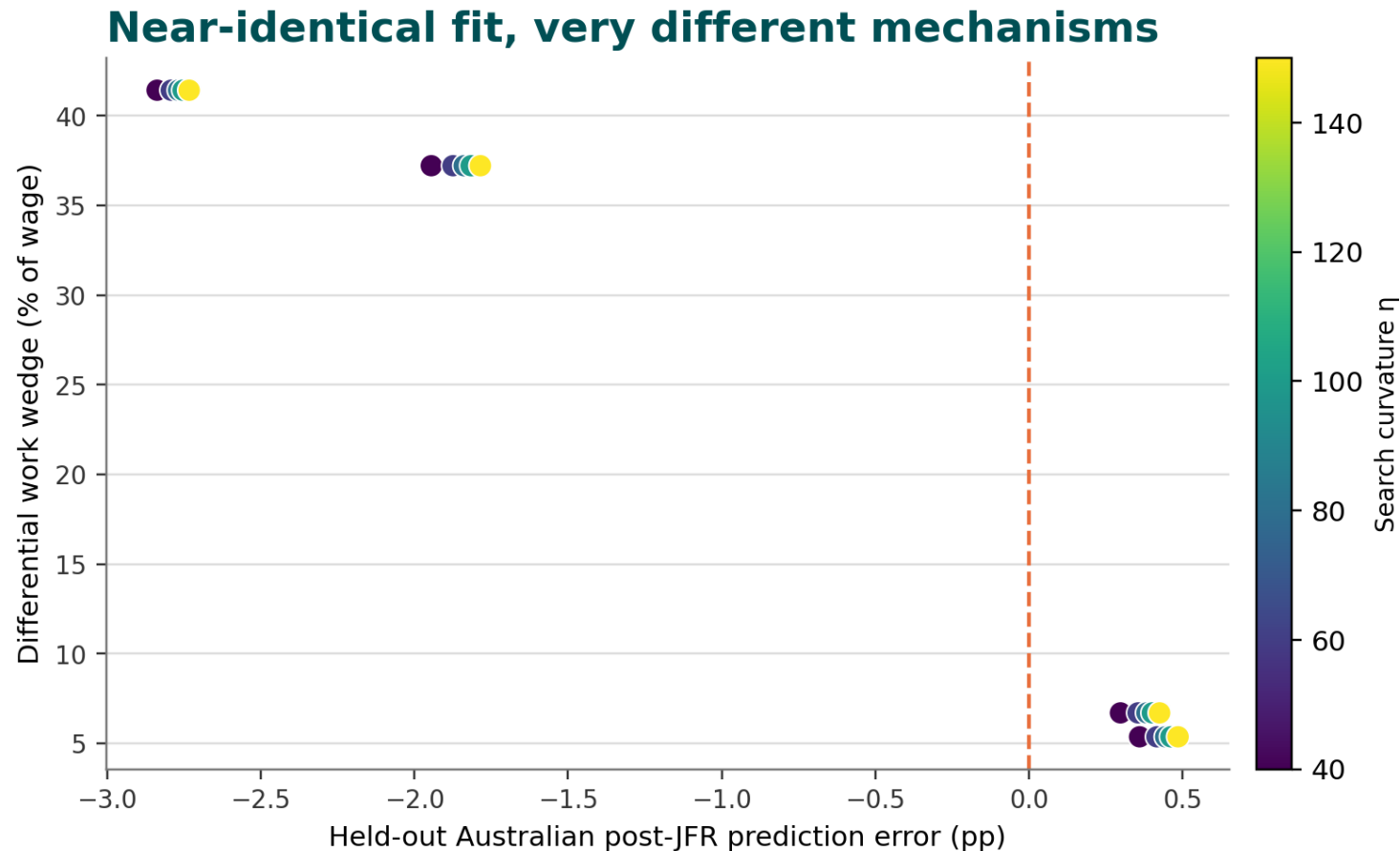
In the benchmark, acceptance—not search—drives most JFR movement



Selected search curvature $\eta = 100$ pins search near one; fixed acceptance thresholds then carry the behavioural response.

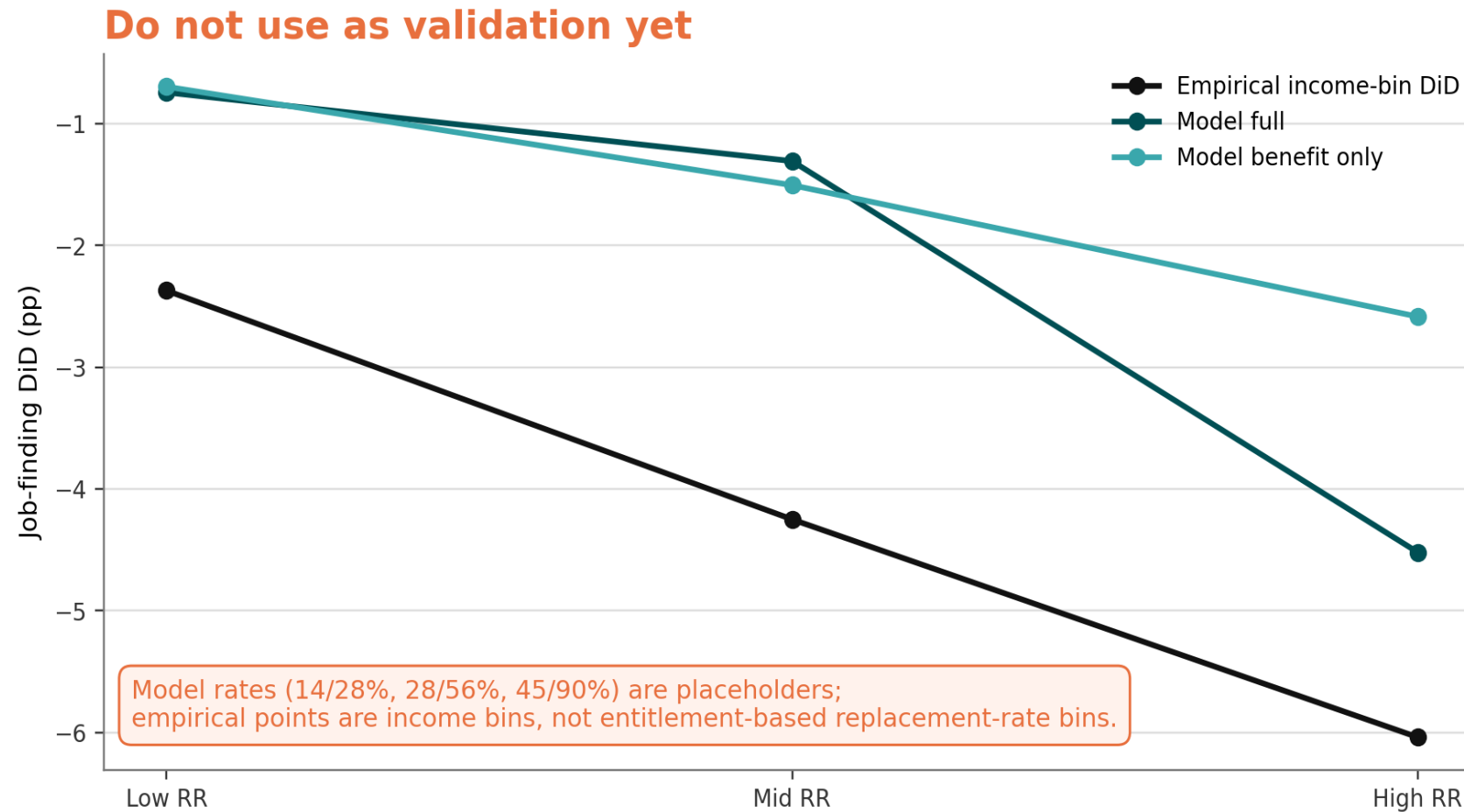
This mechanism is imposed partly by $\eta = 100$ and fixed acceptance parameters; it is not directly validated by applications or offer data.

Weak identification appears in the near-fitting parameter set



Near-fitting set: restricted-grid loss < 0.01. This is functional-form/set-identification uncertainty, not ordinary sampling error. A single «best» grid point is selected by tiny numerical differences in a loss many specifications match almost exactly.

The replacement-rate gradient file is a diagnostic—not a paper result



Rebuild using statutory predicted entitlements and pre-displacement wages before using it as external validation.

04

What is strong, what is fragile?

Separate useful structure from claims the current model cannot carry.

What should be retained

Common surplus index

Benefits and work costs enter the same economic object, creating interpretable nonlinear responses across margins.

Held-out prediction

The restricted calibration reserves Australian post job finding rather than mechanically fitting every target.

Transparent outputs

Self-contained Julia code writes fit, scenario, grid and validation CSVs that can feed a reproducible paper pipeline.

These are solid foundations for an interpretation model once empirical targets, time units and uncertainty are corrected.

What the current aggregate moments cannot identify

Search-and-finding block

- κ level and η curvature
- Matching/contact efficiency μ
- Offer-arrival responsiveness
- Acceptance threshold and dispersion
- Dynamic value surplus

Exit block

- Background firm destruction
- Voluntary versus involuntary exits
- Match-specific quality distribution
- Worker and firm continuation values
- Differential work-surplus wedge

Add external moments or report a set of observationally admissible mechanisms. Six aggregate cells do not identify this many margins.

The residual work wedge needs a neutral label

Current normalisation

Set $h^N = 0$.

Fit common NZ exit increase with ξ .

Fit h^R to the remaining Australian exit increase.

Any R-specific omitted force loads into h^R .

What it may include

Health risk and caregiving

Industry/employer composition

JobKeeper interactions

Benefit administration

Payroll reporting and sample selection

Call $h^R - h^N$ a differential net work-surplus wedge unless independent moments identify health or work disutility.

The current model is not a welfare model

Missing insurance side

- No consumption or saving
- No risk aversion or liquidity constraints
- No household heterogeneity
- No taxes, budget or marginal cost of funds

Missing externalities

- No epidemiological/contact externality
- No congestion or job-rationing externality
- No firm survival or vacancy externality
- No value of improved match quality

Valid conclusion: the paper estimates behavioural transitions and interprets channels. Invalid conclusion: the supplement had positive or negative net social welfare.

Uncertainty must pass through both empirical and structural stages

- Resample the person-week design using blocks and, where relevant, matched pairs.
- Recompute the empirical target vector and covariance matrix in every draw.
- Recalibrate the model; record failures, bound hits and optimiser status.
- For each draw, retain the full economically admissible parameter set—not only one tie-broken optimum.
- Report intervals combining sampling uncertainty with set/functional-form uncertainty.
- Show model fit and counterfactual ranges, not only point bars.

Microdata improve measurement, but the policy still supplies one aggregate shock. Structural precision cannot exceed empirical identification.

05

A submission-ready model roadmap

Make a credible minimal model first; treat richer welfare and equilibrium extensions as optional.

Prioritised structural work before submission

Priority	Action	Why it is binding	Deliverable
1	Recalibrate to redesigned person-week risk-set moments	Current exit targets inherit post-treatment selection	Frozen target table + covariance matrix
2	Correct β , dates and actual benefit entitlements	Current dynamics and replacement rates are not policy-consistent	Date map + entitlement distribution
3	Report an admissible parameter set	Numerical tie-breaking hides weak identification	Ranges for fit, held-out prediction and counterfactuals
4	Complete 2^3 / Shapley decomposition	Current scenarios are non-additive	Mechanism table with interactions
5	Fix firm-exit consistency or simplify matching block	Free entry ignores the model's own exits	Integrated J or exogenous contact interpretation
6	Add numerical tests and reproducible environment	Grid silently omits failures; no project lockfile	Project/Manifest, driver, tests and logs

Stop/go rule: if the redesigned empirical cells are unstable, pause structural refinement and narrow the paper's causal claims.

Minimal model for the main paper

Keep in main text

- Two types and states
- Flow-surplus index
- Search FOC and acceptance
- Reduced-form smooth exit hazard
- Moment-to-parameter table
- Fit + held-out prediction range
- Factorial/Shapley decomposition

Move to appendix

- Bellman derivations
- Full parameter grid
- Optimiser diagnostics
- All shock-path sensitivities
- Alternative normalisations
- Bootstrap/set-construction details
- All subgroup validations

Aim for a transparent quantitative interpretation section, not a second paper inside the empirical paper.

Ambitious extension: explicit match quality and consistent vacancy creation

State extension

Each match draws quality x .
Worker and firm continuation depend on x .
Joint surplus $S_j(x)$ determines an endogenous continuation cutoff x_j^* .
Exit probability is $F(x_j^*)$.

Equilibrium extension

Firm value integrates over type and quality.
Vacancy free entry uses the same exit process.
Benefits affect vacancy value through duration and destruction.
Validate tightness against vacancy data.

This would justify «structural match-quality threshold» language. It is valuable, but not necessary if the current hazard is labelled modestly.

Ambitious extension: insurance, households and public-health externalities

Household block

CRRA utility, assets and borrowing limits
Flat benefits + taper/means tests
Heterogeneous wages and household composition
Consumption response and insurance value

Social block

Benefit financing and marginal cost of funds
Contact intensity by employment state
Epidemiological transmission externality
Vacancy/job-rationing effects
State-contingent optimal benefit paths

This is the route to welfare. It materially expands scope and data requirements; do not let it delay a strong behavioural paper.

Additional moments with the highest identification value

- Applications/search contacts or reported search hours—discipline κ and η .
- Offer receipt, rejection and reservation wages—discipline acceptance threshold and dispersion.
- Vacancies/applications by local market—discipline contact efficiency and tightness.
- Employment-exit reason, stand-down, firm closure and JobKeeper exposure—separate worker continuation from firm destruction.
- Duration dependence in job finding—tests value dynamics and search technology.
- Actual predicted replacement-rate changes—tests the benefit mechanism across policy intensity.
- Post-taper and expiry paths—validate expectations and forward-looking behaviour.

Pre-register which moments are fitted and which are held out. Validation is strongest when the held-out set is chosen before specification search.

How the structural section should be written in the paper

Subsection	Purpose	One main exhibit
Purpose and scope	Interpret transition gaps; explicitly exclude welfare and full GE	One-paragraph mechanism map
Environment	Define types, states, outside options, search, acceptance and exits	State-transition diagram
Equations	Show Bellman equations, search FOC, finding probability and exit hazard	Four-equation box
Identification	Map every fixed/fitted/held-out object to a source or moment	Parameter–moment table
Fit and validation	Show data, benchmark and near-fitting range	Fit + held-out interval figure
Counterfactuals	Report factorial/Shapley effects and underlying margins	Mechanism decomposition
Limits	Partial equilibrium, residual wedge, no welfare, functional-form sensitivity	Short boxed limitations

The benchmark is a visual anchor; conclusions must be based on the admissible range.

Claim discipline: what the model can and cannot support

Defensible

«Model 15 shows that benefits can reduce job finding through surplus and acceptance.»
«Benefits and work-surplus wedges interact in employment exits.»
«A range of mechanisms is consistent with the aggregate moments.»

Not yet defensible

«Most of the response is health rather than benefits.»
«The differential wedge is a health preference.»
«The exit hazard is a structural match-quality cutoff.»
«The supplement increased net welfare.»

Good modelling language distinguishes mechanism possibility, quantitative consistency and identified causal decomposition.

Discussion questions for the research team

- Is the model's central contribution job-finding mechanisms, employment-exit complementarity, or state-contingent policy? Choose one.
- Can the secure data identify applications, offer acceptance, exit reason, JobKeeper exposure or actual entitlement replacement rates?
- Should vacancy creation remain endogenous if its firm value does not share the exit process?
- What economically admissible restrictions rule out 40% wage-equivalent differential wedges?
- Which moments will be genuinely held out before the next calibration?
- Would a simpler reduced-form hazard model tell the same durable story more credibly?
- What quantitative conclusion survives across every near-fitting specification?

Recommended seminar stance: lead with what the model teaches; volunteer the identification limits before the audience does.